PG No. 5 C_{2h} $2/m$ (b-axis setting) [monoclinic] (internal polar octupole)

* Harmonics for rank 0

$\vec{\mathbb{Q}}_0^{(3,3)}[q](A_g)$

** symmetry

1

** expression

$$\frac{\sqrt{105}Q_{xyz}xyz}{7}+\frac{\sqrt{7}Q_x^\alpha x\left(2x^2-3y^2-3z^2\right)}{14}+\frac{\sqrt{105}Q_x^\beta x\left(y-z\right)\left(y+z\right)}{14}-\frac{\sqrt{7}Q_y^\alpha y\left(3x^2-2y^2+3z^2\right)}{14}$$
$$-\frac{\sqrt{105}Q_y^\beta y\left(x-z\right)\left(x+z\right)}{14}-\frac{\sqrt{7}Q_z^\alpha z\left(3x^2+3y^2-2z^2\right)}{14}+\frac{\sqrt{105}Q_z^\beta z\left(x-y\right)\left(x+y\right)}{14}$$

* Harmonics for rank 1

$\vec{\mathbb{G}}_1^{(3,2)}[q](A_g)$

** symmetry

y

** expression

$$\frac{\sqrt{105}Q_{xyz}y\left(x-z\right)\left(x+z\right)}{14}-\frac{3\sqrt{7}Q_x^\alpha z\left(4x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}Q_x^\beta z\left(2x^2+y^2-z^2\right)}{28}$$
$$+\frac{\sqrt{105}Q_y^\beta xyz}{7}-\frac{3\sqrt{7}Q_z^\alpha x\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{105}Q_z^\beta x\left(x^2-y^2-2z^2\right)}{28}$$

$\vec{\mathbb{G}}_1^{(3,2)}[q](B_g,1)$

** symmetry

x

** expression

$$-\frac{\sqrt{105}Q_{xyz}x\left(y-z\right)\left(y+z\right)}{14}+\frac{\sqrt{105}Q_x^\beta xyz}{7}-\frac{3\sqrt{7}Q_y^\alpha z\left(x^2-4y^2+z^2\right)}{28}$$
$$-\frac{\sqrt{105}Q_y^\beta z\left(x^2+2y^2-z^2\right)}{28}+\frac{3\sqrt{7}Q_z^\alpha y\left(x^2+y^2-4z^2\right)}{28}-\frac{\sqrt{105}Q_z^\beta y\left(x^2-y^2+2z^2\right)}{28}$$

$\vec{\mathbb{G}}_1^{(3,2)}[q](B_g,2)$

** symmetry

z

** expression

$$-\frac{\sqrt{105}Q_{xyz}z\left(x-y\right)\left(x+y\right)}{14}+\frac{3\sqrt{7}Q_x^\alpha y\left(4x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}Q_x^\beta y\left(2x^2-y^2+z^2\right)}{28}$$
$$+\frac{3\sqrt{7}Q_y^\alpha x\left(x^2-4y^2+z^2\right)}{28}+\frac{\sqrt{105}Q_y^\beta x\left(x^2-2y^2-z^2\right)}{28}+\frac{\sqrt{105}Q_z^\beta xyz}{7}$$

$\vec{\mathbb{Q}}_1^{(3,1)}[q](A_u)$

** symmetry

y

** expression

$$\frac{\sqrt{21}Q_{xyz}xz}{7}-\frac{3\sqrt{35}Q_x^\alpha xy}{35}+\frac{\sqrt{21}Q_x^\beta xy}{7}-\frac{3\sqrt{35}Q_y^\alpha\left(x^2-2y^2+z^2\right)}{70}-\frac{\sqrt{21}Q_y^\beta\left(x-z\right)\left(x+z\right)}{14}-\frac{3\sqrt{35}Q_z^\alpha yz}{35}-\frac{\sqrt{21}Q_z^\beta yz}{7}$$

$\vec{\mathbb{Q}}_1^{(3,3)}[q](A_u)$

** symmetry

y

** expression

$$-\frac{\sqrt{35}Q_{xyz}xz\left(x^2-6y^2+z^2\right)}{14}+\frac{5\sqrt{21}Q_x^\alpha xy\left(4x^2-3y^2-3z^2\right)}{84}-\frac{\sqrt{35}Q_x^\beta xy\left(2x^2-5y^2+9z^2\right)}{28}$$
$$+\frac{\sqrt{21}Q_y^\alpha\left(3x^4-24x^2y^2+6x^2z^2+8y^4-24y^2z^2+3z^4\right)}{84}+\frac{\sqrt{35}Q_y^\beta\left(x-z\right)\left(x+z\right)\left(x^2-6y^2+z^2\right)}{28}$$
$$-\frac{5\sqrt{21}Q_z^\alpha yz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}Q_z^\beta yz\left(9x^2-5y^2+2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_1^{(3,1)}[q](B_u,1)$

** symmetry

x

** expression

$$\frac{\sqrt{21}Q_{xyz}yz}{7}+\frac{3\sqrt{35}Q_x^{\alpha}\left(2x^2-y^2-z^2\right)}{70}+\frac{\sqrt{21}Q_x^{\beta}\left(y-z\right)\left(y+z\right)}{14}-\frac{3\sqrt{35}Q_y^{\alpha}xy}{35}-\frac{\sqrt{21}Q_y^{\beta}xy}{7}-\frac{3\sqrt{35}Q_z^{\alpha}xz}{35}+\frac{\sqrt{21}Q_z^{\beta}xz}{7}$$

$\vec{\mathbb{Q}}_1^{(3,1)}[q](B_u,2)$

** symmetry

z

** expression

$$\frac{\sqrt{21}Q_{xyz}xy}{7}-\frac{3\sqrt{35}Q_x^{\alpha}xz}{35}-\frac{\sqrt{21}Q_x^{\beta}xz}{7}-\frac{3\sqrt{35}Q_y^{\alpha}yz}{35}+\frac{\sqrt{21}Q_y^{\beta}yz}{7}-\frac{3\sqrt{35}Q_z^{\alpha}\left(x^2+y^2-2z^2\right)}{70}+\frac{\sqrt{21}Q_z^{\beta}\left(x-y\right)\left(x+y\right)}{14}$$

$\vec{\mathbb{Q}}_1^{(3,3)}[q](B_u,1)$

** symmetry

x

** expression

$$\frac{\sqrt{35}Q_{xyz}yz\left(6x^2-y^2-z^2\right)}{14}+\frac{\sqrt{21}Q_x^{\alpha}\left(8x^4-24x^2y^2-24x^2z^2+3y^4+6y^2z^2+3z^4\right)}{84}+\frac{\sqrt{35}Q_x^{\beta}\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{28}$$
$$-\frac{5\sqrt{21}Q_y^{\alpha}xy\left(3x^2-4y^2+3z^2\right)}{84}-\frac{\sqrt{35}Q_y^{\beta}xy\left(5x^2-2y^2-9z^2\right)}{28}-\frac{5\sqrt{21}Q_z^{\alpha}xz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}Q_z^{\beta}xz\left(5x^2-9y^2-2z^2\right)}{28}$$

$\vec{\mathbb{Q}}_1^{(3,3)}[q](B_u,2)$

** symmetry

z

** expression

$$-\frac{\sqrt{35}Q_{xyz}xy\left(x^2+y^2-6z^2\right)}{14}+\frac{5\sqrt{21}Q_x^{\alpha}xz\left(4x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{35}Q_x^{\beta}xz\left(2x^2+9y^2-5z^2\right)}{28}-\frac{5\sqrt{21}Q_y^{\alpha}yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$-\frac{\sqrt{35}Q_y^{\beta}yz\left(9x^2+2y^2-5z^2\right)}{28}+\frac{\sqrt{21}Q_z^{\alpha}\left(3x^4+6x^2y^2-24x^2z^2+3y^4-24y^2z^2+8z^4\right)}{84}-\frac{\sqrt{35}Q_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{28}$$

* Harmonics for rank 2

$\vec{\mathbb{Q}}_2^{(3,-1)}[q](A_g,1)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}Q_x^{\alpha}x}{14}-\frac{\sqrt{35}Q_x^{\beta}x}{14}-\frac{\sqrt{21}Q_y^{\alpha}y}{14}+\frac{\sqrt{35}Q_y^{\beta}y}{14}+\frac{\sqrt{21}Q_z^{\alpha}z}{7}$$

$\vec{\mathbb{Q}}_2^{(3,-1)}[q](A_g,2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{7}Q_x^{\alpha}x}{14}-\frac{\sqrt{105}Q_x^{\beta}x}{42}-\frac{3\sqrt{7}Q_y^{\alpha}y}{14}-\frac{\sqrt{105}Q_y^{\beta}y}{42}+\frac{\sqrt{105}Q_z^{\beta}z}{21}$$

$\vec{\mathbb{Q}}_2^{(3,-1)}[q](A_g,3)$

** symmetry

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{105}Q_{xyz}y}{21}-\frac{\sqrt{7}Q_x^{\alpha}z}{7}-\frac{\sqrt{105}Q_x^{\beta}z}{21}-\frac{\sqrt{7}Q_z^{\alpha}x}{7}+\frac{\sqrt{105}Q_z^{\beta}x}{21}$$

$\vec{\mathbb{Q}}_2^{(3,1)}[q](A_g,1)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{21}Q_x^{\alpha}x\left(4x^2-21y^2+9z^2\right)}{84}+\frac{\sqrt{35}Q_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{28}+\frac{\sqrt{21}Q_y^{\alpha}y\left(21x^2-4y^2-9z^2\right)}{84}$$
$$+\frac{\sqrt{35}Q_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{28}-\frac{\sqrt{21}Q_z^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{21}$$

$\vec{\mathbb{Q}}_2^{(3,1)}[q](A_g,2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{7}Q_x^{\alpha}x\left(4x^2-y^2-11z^2\right)}{28}+\frac{\sqrt{105}Q_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{84}+\frac{\sqrt{7}Q_y^{\alpha}y\left(x^2-4y^2+11z^2\right)}{28}$$
$$-\frac{\sqrt{105}Q_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{84}-\frac{5\sqrt{7}Q_z^{\alpha}z\left(x-y\right)\left(x+y\right)}{14}+\frac{\sqrt{105}Q_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{42}$$

$\vec{\mathbb{Q}}_2^{(3,1)}[q](A_g,3)$

** symmetry

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{105}Q_{xyz}y\left(3x^2-2y^2+3z^2\right)}{42}+\frac{\sqrt{7}Q_x^{\alpha}z\left(4x^2-y^2-z^2\right)}{28}-\frac{\sqrt{105}Q_x^{\beta}z\left(6x^2-9y^2+z^2\right)}{84}$$
$$-\frac{5\sqrt{7}Q_y^{\alpha}xyz}{7}-\frac{\sqrt{7}Q_z^{\alpha}x\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{105}Q_z^{\beta}x\left(x^2-9y^2+6z^2\right)}{84}$$

$\vec{\mathbb{Q}}_2^{(3,3)}[q](A_g,1)$

** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{3\sqrt{770}Q_{xyz}xyz\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{462}Q_x^{\alpha}x\left(4x^4+x^2y^2-41x^2z^2-3y^4+15y^2z^2+18z^4\right)}{1848}$$
$$-\frac{\sqrt{770}Q_x^{\beta}x\left(2x^4+11x^2y^2-31x^2z^2+9y^4-87y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}Q_y^{\alpha}y\left(3x^4-x^2y^2-15x^2z^2-4y^4+41y^2z^2-18z^4\right)}{1848}+\frac{\sqrt{770}Q_y^{\beta}y\left(9x^4+11x^2y^2-87x^2z^2+2y^4-31y^2z^2+30z^4\right)}{616}$$
$$+\frac{5\sqrt{462}Q_z^{\alpha}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{1848}-\frac{3\sqrt{770}Q_z^{\beta}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$

$\vec{\mathbb{Q}}_2^{(3,3)}[q](A_g,2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2}-\frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{3\sqrt{2310}Q_{xyz}xyz\left(x-y\right)\left(x+y\right)}{44}+\frac{5\sqrt{154}Q_x^{\alpha}x\left(4x^4-27x^2y^2-13x^2z^2+11y^4+15y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}Q_x^{\beta}x\left(2x^4-73x^2y^2+53x^2z^2+51y^4-87y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}Q_y^{\alpha}y\left(11x^4-27x^2y^2+15x^2z^2+4y^4-13y^2z^2+4z^4\right)}{616}$$
$$-\frac{\sqrt{2310}Q_y^{\beta}y\left(51x^4-73x^2y^2-87x^2z^2+2y^4+53y^2z^2-12z^4\right)}{1848}-\frac{5\sqrt{154}Q_z^{\alpha}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{88}$$
$$+\frac{\sqrt{2310}Q_z^{\beta}z\left(39x^4-174x^2y^2-20x^2z^2+39y^4-20y^2z^2+4z^4\right)}{1848}$$

$\vec{\mathbb{Q}}_2^{(3,3)}[q](A_g,3)$

** symmetry

$$\sqrt{3}xz$$

** expression

$$-\frac{\sqrt{2310}Q_{xyz}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{462}+\frac{5\sqrt{154}Q_x^{\alpha}z\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{2310}Q_x^{\beta}z\left(12x^4+66x^2y^2-46x^2z^2-9y^4-4y^2z^2+5z^4\right)}{924}-\frac{5\sqrt{154}Q_y^{\alpha}xyz\left(x^2-2y^2+z^2\right)}{44}-\frac{3\sqrt{2310}Q_y^{\beta}xyz\left(x-z\right)\left(x+z\right)}{44}$$
$$+\frac{5\sqrt{154}Q_z^{\alpha}x\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}-\frac{\sqrt{2310}Q_z^{\beta}x\left(5x^4-4x^2y^2-46x^2z^2-9y^4+66y^2z^2+12z^4\right)}{924}$$

$\vec{\mathbb{Q}}_2^{(3,-1)}[q](B_g,1)$

** symmetry

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{105}Q_{xyz}x}{21}-\frac{\sqrt{7}Q_y^\alpha z}{7}+\frac{\sqrt{105}Q_y^\beta z}{21}-\frac{\sqrt{7}Q_z^\alpha y}{7}-\frac{\sqrt{105}Q_z^\beta y}{21}$$

$\vec{\mathbb{Q}}_2^{(3,-1)}[q](B_g,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}Q_{xyz}z}{21}-\frac{\sqrt{7}Q_x^\alpha y}{7}+\frac{\sqrt{105}Q_x^\beta y}{21}-\frac{\sqrt{7}Q_y^\alpha x}{7}-\frac{\sqrt{105}Q_y^\beta x}{21}$$

$\vec{\mathbb{Q}}_2^{(3,1)}[q](B_g,1)$
** symmetry

$$\sqrt{3}yz$$

** expression

$$-\frac{\sqrt{105}Q_{xyz}x\left(2x^2-3y^2-3z^2\right)}{42}-\frac{5\sqrt{7}Q_x^\alpha xyz}{7}-\frac{\sqrt{7}Q_y^\alpha z\left(x^2-4y^2+z^2\right)}{28}$$
$$-\frac{\sqrt{105}Q_y^\beta z\left(9x^2-6y^2-z^2\right)}{84}-\frac{\sqrt{7}Q_z^\alpha y\left(x^2+y^2-4z^2\right)}{28}+\frac{\sqrt{105}Q_z^\beta y\left(9x^2-y^2-6z^2\right)}{84}$$

$\vec{\mathbb{Q}}_2^{(3,1)}[q](B_g,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{105}Q_{xyz}z\left(3x^2+3y^2-2z^2\right)}{42}+\frac{\sqrt{7}Q_x^\alpha y\left(4x^2-y^2-z^2\right)}{28}+\frac{\sqrt{105}Q_x^\beta y\left(6x^2+y^2-9z^2\right)}{84}$$
$$-\frac{\sqrt{7}Q_y^\alpha x\left(x^2-4y^2+z^2\right)}{28}-\frac{\sqrt{105}Q_y^\beta x\left(x^2+6y^2-9z^2\right)}{84}-\frac{5\sqrt{7}Q_z^\alpha xyz}{7}$$

$\vec{\mathbb{Q}}_2^{(3,3)}[q](B_g,1)$
** symmetry

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{2310}Q_{xyz}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{462}+\frac{5\sqrt{154}Q_x^\alpha xyz\left(2x^2-y^2-z^2\right)}{44}+\frac{3\sqrt{2310}Q_x^\beta xyz\left(y-z\right)\left(y+z\right)}{44}$$
$$+\frac{5\sqrt{154}Q_y^\alpha z\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{308}+\frac{\sqrt{2310}Q_y^\beta z\left(9x^4-66x^2y^2+4x^2z^2-12y^4+46y^2z^2-5z^4\right)}{924}$$
$$+\frac{5\sqrt{154}Q_z^\alpha y\left(x^4+2x^2y^2-12x^2z^2+y^4-12y^2z^2+8z^4\right)}{308}-\frac{\sqrt{2310}Q_z^\beta y\left(9x^4+4x^2y^2-66x^2z^2-5y^4+46y^2z^2-12z^4\right)}{924}$$

$\vec{\mathbb{Q}}_2^{(3,3)}[q](B_g,2)$
** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{2310}Q_{xyz}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{462}+\frac{5\sqrt{154}Q_x^\alpha y\left(8x^4-12x^2y^2-12x^2z^2+y^4+2y^2z^2+z^4\right)}{308}$$
$$-\frac{\sqrt{2310}Q_x^\beta y\left(12x^4-46x^2y^2+66x^2z^2+5y^4-4y^2z^2-9z^4\right)}{924}+\frac{5\sqrt{154}Q_y^\alpha x\left(x^4-12x^2y^2+2x^2z^2+8y^4-12y^2z^2+z^4\right)}{308}$$
$$+\frac{\sqrt{2310}Q_y^\beta x\left(5x^4-46x^2y^2-4x^2z^2+12y^4+66y^2z^2-9z^4\right)}{924}-\frac{5\sqrt{154}Q_z^\alpha xyz\left(x^2+y^2-2z^2\right)}{44}+\frac{3\sqrt{2310}Q_z^\beta xyz\left(x-y\right)\left(x+y\right)}{44}$$

$\vec{\mathbb{G}}_2^{(3,0)}[q](A_u,1)$
** symmetry

$$-\frac{x^2}{2}-\frac{y^2}{2}+z^2$$

** expression

$$-\frac{\sqrt{210}Q_{xyz}\left(x-y\right)\left(x+y\right)}{28}-\frac{3\sqrt{14}Q_x^\alpha yz}{28}-\frac{\sqrt{210}Q_x^\beta yz}{28}+\frac{3\sqrt{14}Q_y^\alpha xz}{28}-\frac{\sqrt{210}Q_y^\beta xz}{28}+\frac{\sqrt{210}Q_z^\beta xy}{14}$$

$\vec{\mathbb{G}}_2^{(3,0)}[q](A_u, 2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$-\frac{\sqrt{70}Q_{xyz}\left(x^2+y^2-2z^2\right)}{28} - \frac{\sqrt{42}Q_x^{\alpha}yz}{28} + \frac{3\sqrt{70}Q_x^{\beta}yz}{28} - \frac{\sqrt{42}Q_y^{\alpha}xz}{28} - \frac{3\sqrt{70}Q_y^{\beta}xz}{28} + \frac{\sqrt{42}Q_z^{\alpha}xy}{14}$$

$\vec{\mathbb{G}}_2^{(3,0)}[q](A_u, 3)$

** symmetry

$$\sqrt{3}xz$$

** expression

$$\frac{\sqrt{42}Q_x^{\alpha}xy}{7} + \frac{\sqrt{42}Q_y^{\alpha}(x-z)(x+z)}{28} + \frac{\sqrt{70}Q_y^{\beta}\left(x^2-2y^2+z^2\right)}{28} - \frac{\sqrt{42}Q_z^{\alpha}yz}{7}$$

$\vec{\mathbb{G}}_2^{(3,2)}[q](A_u, 1)$

** symmetry

$$-\frac{x^2}{2} - \frac{y^2}{2} + z^2$$

** expression

$$\frac{\sqrt{105}Q_{xyz}(x-y)(x+y)\left(x^2+y^2-6z^2\right)}{42} + \frac{5\sqrt{7}Q_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{28} - \frac{\sqrt{105}Q_x^{\beta}yz\left(12x^2-9y^2+5z^2\right)}{84}$$
$$+ \frac{5\sqrt{7}Q_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{28} + \frac{\sqrt{105}Q_y^{\beta}xz\left(9x^2-12y^2-5z^2\right)}{84} - \frac{\sqrt{105}Q_z^{\beta}xy\left(x^2+y^2-6z^2\right)}{21}$$

$\vec{\mathbb{G}}_2^{(3,2)}[q](A_u, 2)$

** symmetry

$$\frac{\sqrt{3}x^2}{2} - \frac{\sqrt{3}y^2}{2}$$

** expression

$$\frac{\sqrt{35}Q_{xyz}\left(x^4-12x^2y^2+6x^2z^2+y^4+6y^2z^2-2z^4\right)}{42} + \frac{5\sqrt{21}Q_x^{\alpha}yz\left(6x^2-y^2-z^2\right)}{84} + \frac{\sqrt{35}Q_x^{\beta}yz\left(36x^2+y^2-13z^2\right)}{84}$$
$$- \frac{5\sqrt{21}Q_y^{\alpha}xz\left(x^2-6y^2+z^2\right)}{84} - \frac{\sqrt{35}Q_y^{\beta}xz\left(x^2+36y^2-13z^2\right)}{84} + \frac{5\sqrt{21}Q_z^{\alpha}xy\left(x^2+y^2-6z^2\right)}{42} - \frac{\sqrt{35}Q_z^{\beta}xy(x-y)(x+y)}{6}$$

$\vec{\mathbb{G}}_2^{(3,2)}[q](A_u, 3)$

** symmetry

$$\sqrt{3}xz$$

** expression

$$-\frac{\sqrt{35}Q_{xyz}xz(x-z)(x+z)}{6} + \frac{5\sqrt{21}Q_x^{\alpha}xy\left(4x^2-3y^2-3z^2\right)}{84} - \frac{\sqrt{35}Q_x^{\beta}xy\left(2x^2-y^2-3z^2\right)}{12} + \frac{5\sqrt{21}Q_y^{\alpha}(x-z)(x+z)\left(x^2-6y^2+z^2\right)}{84}$$
$$+ \frac{\sqrt{35}Q_y^{\beta}\left(5x^4-12x^2y^2-18x^2z^2+4y^4-12y^2z^2+5z^4\right)}{84} + \frac{5\sqrt{21}Q_z^{\alpha}yz\left(3x^2+3y^2-4z^2\right)}{84} + \frac{\sqrt{35}Q_z^{\beta}yz\left(3x^2+y^2-2z^2\right)}{12}$$

$\vec{\mathbb{G}}_2^{(3,0)}[q](B_u, 1)$

** symmetry

$$\sqrt{3}yz$$

** expression

$$-\frac{\sqrt{42}Q_x^{\alpha}(y-z)(y+z)}{28} - \frac{\sqrt{70}Q_x^{\beta}\left(2x^2-y^2-z^2\right)}{28} - \frac{\sqrt{42}Q_y^{\alpha}xy}{7} + \frac{\sqrt{42}Q_z^{\alpha}xz}{7}$$

$\vec{\mathbb{G}}_2^{(3,0)}[q](B_u, 2)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$-\frac{\sqrt{42}Q_x^{\alpha}xz}{7} + \frac{\sqrt{42}Q_y^{\alpha}yz}{7} - \frac{\sqrt{42}Q_z^{\alpha}(x-y)(x+y)}{28} + \frac{\sqrt{70}Q_z^{\beta}\left(x^2+y^2-2z^2\right)}{28}$$

$\vec{\mathbb{G}}_2^{(3,2)}[q](B_u, 1)$

** symmetry

$$\sqrt{3}yz$$

** expression

$$\frac{\sqrt{35}Q_{xyz}yz\left(y-z\right)\left(y+z\right)}{6}+\frac{5\sqrt{21}Q_x^{\alpha}\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{84}+\frac{\sqrt{35}Q_x^{\beta}\left(4x^4-12x^2y^2-12x^2z^2+5y^4-18y^2z^2+5z^4\right)}{84}$$
$$+\frac{5\sqrt{21}Q_y^{\alpha}xy\left(3x^2-4y^2+3z^2\right)}{84}+\frac{\sqrt{35}Q_y^{\beta}xy\left(x^2-2y^2+3z^2\right)}{12}-\frac{5\sqrt{21}Q_z^{\alpha}xz\left(3x^2+3y^2-4z^2\right)}{84}+\frac{\sqrt{35}Q_z^{\beta}xz\left(x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{G}}_2^{(3,2)}[q](B_u, 2)$

** symmetry

$$\sqrt{3}xy$$

** expression

$$\frac{\sqrt{35}Q_{xyz}xy\left(x-y\right)\left(x+y\right)}{6}-\frac{5\sqrt{21}Q_x^{\alpha}xz\left(4x^2-3y^2-3z^2\right)}{84}-\frac{\sqrt{35}Q_x^{\beta}xz\left(2x^2-3y^2-z^2\right)}{12}-\frac{5\sqrt{21}Q_y^{\alpha}yz\left(3x^2-4y^2+3z^2\right)}{84}$$
$$+\frac{\sqrt{35}Q_y^{\beta}yz\left(3x^2-2y^2+z^2\right)}{12}-\frac{5\sqrt{21}Q_z^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{84}+\frac{\sqrt{35}Q_z^{\beta}\left(5x^4-18x^2y^2-12x^2z^2+5y^4-12y^2z^2+4z^4\right)}{84}$$

* Harmonics for rank 3

$\vec{\mathbb{G}}_3^{(3,-2)}[q](A_g, 1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{\sqrt{3}Q_x^{\beta}x}{3}-\frac{\sqrt{3}Q_y^{\beta}y}{3}-\frac{\sqrt{3}Q_z^{\beta}z}{3}$$

$\vec{\mathbb{G}}_3^{(3,-2)}[q](A_g, 2)$

** symmetry

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$\frac{\sqrt{3}Q_x^{\alpha}z}{4}-\frac{\sqrt{5}Q_x^{\beta}z}{4}-\frac{\sqrt{3}Q_z^{\alpha}x}{4}-\frac{\sqrt{5}Q_z^{\beta}x}{4}$$

$\vec{\mathbb{G}}_3^{(3,-2)}[q](A_g, 3)$

** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{\sqrt{3}Q_{xyz}y}{3}+\frac{\sqrt{5}Q_x^{\alpha}z}{4}-\frac{\sqrt{3}Q_x^{\beta}z}{12}+\frac{\sqrt{5}Q_z^{\alpha}x}{4}+\frac{\sqrt{3}Q_z^{\beta}x}{12}$$

$\vec{\mathbb{G}}_3^{(3,0)}[q](A_g, 1)$

** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{10}Q_x^{\alpha}x\left(y-z\right)\left(y+z\right)}{4}-\frac{\sqrt{6}Q_x^{\beta}x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}Q_y^{\alpha}y\left(x-z\right)\left(x+z\right)}{4}$$
$$+\frac{\sqrt{6}Q_y^{\beta}y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}Q_z^{\alpha}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{6}Q_z^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}$$

$\vec{\mathbb{G}}_3^{(3,0)}[q](A_g, 2)$

** symmetry

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$\frac{\sqrt{10}Q_{xyz}y\left(x-z\right)\left(x+z\right)}{4}+\frac{\sqrt{6}Q_x^{\alpha}z\left(3x^2+3y^2-2z^2\right)}{12}+\frac{\sqrt{10}Q_x^{\beta}z\left(x-y\right)\left(x+y\right)}{4}+\frac{\sqrt{10}Q_y^{\beta}xyz}{2}+\frac{\sqrt{6}Q_z^{\alpha}x\left(2x^2-3y^2-3z^2\right)}{12}-\frac{\sqrt{10}Q_z^{\beta}x\left(y-z\right)\left(y+z\right)}{4}$$

$\vec{\mathbb{G}}_3^{(3,0)}[q](A_g,3)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{\sqrt{6}Q_{xyz}y\left(3x^2-2y^2+3z^2\right)}{12}+\frac{\sqrt{10}Q_x^{\alpha}z\left(x-y\right)\left(x+y\right)}{4}-\frac{\sqrt{6}Q_x^{\beta}z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$-\frac{\sqrt{10}Q_y^{\alpha}xyz}{2}-\frac{\sqrt{10}Q_z^{\alpha}x\left(y-z\right)\left(y+z\right)}{4}-\frac{\sqrt{6}Q_z^{\beta}x\left(2x^2-3y^2-3z^2\right)}{12}$$

$\vec{\mathbb{G}}_3^{(3,2)}[q](A_g,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{7\sqrt{55}Q_x^{\alpha}x\left(y-z\right)\left(y+z\right)\left(2x^2-y^2-z^2\right)}{44}+\frac{\sqrt{33}Q_x^{\beta}x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{132}$$
$$+\frac{7\sqrt{55}Q_y^{\alpha}y\left(x-z\right)\left(x+z\right)\left(x^2-2y^2+z^2\right)}{44}+\frac{\sqrt{33}Q_y^{\beta}y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{132}$$
$$-\frac{7\sqrt{55}Q_z^{\alpha}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44}+\frac{\sqrt{33}Q_z^{\beta}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}$$

$\vec{\mathbb{G}}_3^{(3,2)}[q](A_g,2)$
** symmetry

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$-\frac{7\sqrt{55}Q_{xyz}y\left(x-z\right)\left(x+z\right)\left(x^2-2y^2+z^2\right)}{44}+\frac{5\sqrt{33}Q_x^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{264}$$
$$+\frac{\sqrt{55}Q_x^{\beta}z\left(4x^4-27x^2y^2+x^2z^2-10y^4+29y^2z^2-3z^4\right)}{88}-\frac{7\sqrt{55}Q_y^{\beta}xyz\left(x^2-2y^2+z^2\right)}{22}$$
$$+\frac{5\sqrt{33}Q_z^{\alpha}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{264}-\frac{\sqrt{55}Q_z^{\beta}x\left(3x^4-29x^2y^2-x^2z^2+10y^4+27y^2z^2-4z^4\right)}{88}$$

$\vec{\mathbb{G}}_3^{(3,2)}[q](A_g,3)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{\sqrt{33}Q_{xyz}y\left(15x^4-40x^2y^2+30x^2z^2+8y^4-40y^2z^2+15z^4\right)}{132}+\frac{\sqrt{55}Q_x^{\alpha}z\left(10x^4+27x^2y^2-29x^2z^2-4y^4-y^2z^2+3z^4\right)}{88}$$
$$+\frac{\sqrt{33}Q_x^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{264}+\frac{7\sqrt{55}Q_y^{\alpha}xyz\left(x^2-2y^2+z^2\right)}{22}$$
$$+\frac{\sqrt{55}Q_z^{\alpha}x\left(3x^4-x^2y^2-29x^2z^2-4y^4+27y^2z^2+10z^4\right)}{88}-\frac{\sqrt{33}Q_z^{\beta}x\left(11x^4-55x^2y^2-55x^2z^2+60y^4-195y^2z^2+60z^4\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,-2)}[q](B_g,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

** expression

$$-\frac{\sqrt{3}Q_y^{\alpha}z}{4}-\frac{\sqrt{5}Q_y^{\beta}z}{4}+\frac{\sqrt{3}Q_z^{\alpha}y}{4}-\frac{\sqrt{5}Q_z^{\beta}y}{4}$$

$\vec{\mathbb{G}}_3^{(3,-2)}[q](B_g,2)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$-\frac{\sqrt{3}Q_x^{\alpha}y}{4}-\frac{\sqrt{5}Q_x^{\beta}y}{4}+\frac{\sqrt{3}Q_y^{\alpha}x}{4}-\frac{\sqrt{5}Q_y^{\beta}x}{4}$$

$\vec{\mathbb{G}}_3^{(3,-2)}[q](B_g, 3)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

** expression

$$\frac{\sqrt{3}Q_{xyz}x}{3} + \frac{\sqrt{5}Q_y^\alpha z}{4} + \frac{\sqrt{3}Q_y^\beta z}{12} + \frac{\sqrt{5}Q_z^\alpha y}{4} - \frac{\sqrt{3}Q_z^\beta y}{12}$$

$\vec{\mathbb{G}}_3^{(3,-2)}[q](B_g, 4)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{\sqrt{3}Q_{xyz}z}{3} + \frac{\sqrt{5}Q_x^\alpha y}{4} + \frac{\sqrt{3}Q_x^\beta y}{12} + \frac{\sqrt{5}Q_y^\alpha x}{4} - \frac{\sqrt{3}Q_y^\beta x}{12}$$

$\vec{\mathbb{G}}_3^{(3,0)}[q](B_g, 1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$-\frac{\sqrt{10}Q_{xyz}x\,(y-z)\,(y+z)}{4} + \frac{\sqrt{10}Q_x^\beta xyz}{2} - \frac{\sqrt{6}Q_y^\alpha z\left(3x^2+3y^2-2z^2\right)}{12}$$
$$-\frac{\sqrt{10}Q_y^\beta z\,(x-y)\,(x+y)}{4} + \frac{\sqrt{6}Q_z^\alpha y\left(3x^2-2y^2+3z^2\right)}{12} - \frac{\sqrt{10}Q_z^\beta y\,(x-z)\,(x+z)}{4}$$

$\vec{\mathbb{G}}_3^{(3,0)}[q](B_g, 2)$

** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-\frac{\sqrt{10}Q_{xyz}z\,(x-y)\,(x+y)}{4} - \frac{\sqrt{6}Q_x^\alpha y\left(3x^2-2y^2+3z^2\right)}{12} + \frac{\sqrt{10}Q_x^\beta y\,(x-z)\,(x+z)}{4}$$
$$-\frac{\sqrt{6}Q_y^\alpha x\left(2x^2-3y^2-3z^2\right)}{12} + \frac{\sqrt{10}Q_y^\beta x\,(y-z)\,(y+z)}{4} + \frac{\sqrt{10}Q_z^\beta xyz}{2}$$

$\vec{\mathbb{G}}_3^{(3,0)}[q](B_g, 3)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

** expression

$$\frac{\sqrt{6}Q_{xyz}x\left(2x^2-3y^2-3z^2\right)}{12} - \frac{\sqrt{10}Q_x^\alpha xyz}{2} - \frac{\sqrt{10}Q_y^\alpha z\,(x-y)\,(x+y)}{4}$$
$$+\frac{\sqrt{6}Q_y^\beta z\left(3x^2+3y^2-2z^2\right)}{12} - \frac{\sqrt{10}Q_z^\alpha y\,(x-z)\,(x+z)}{4} - \frac{\sqrt{6}Q_z^\beta y\left(3x^2-2y^2+3z^2\right)}{12}$$

$\vec{\mathbb{G}}_3^{(3,0)}[q](B_g, 4)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{6}Q_{xyz}z\left(3x^2+3y^2-2z^2\right)}{12} + \frac{\sqrt{10}Q_x^\alpha y\,(x-z)\,(x+z)}{4} + \frac{\sqrt{6}Q_x^\beta y\left(3x^2-2y^2+3z^2\right)}{12}$$
$$+\frac{\sqrt{10}Q_y^\alpha x\,(y-z)\,(y+z)}{4} + \frac{\sqrt{6}Q_y^\beta x\left(2x^2-3y^2-3z^2\right)}{12} - \frac{\sqrt{10}Q_z^\alpha xyz}{2}$$

$\vec{\mathbb{G}}_3^{(3,2)}[q](B_g, 1)$

** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$-\frac{7\sqrt{55}Q_{xyz}x\left(y-z\right)\left(y+z\right)\left(2x^2-y^2-z^2\right)}{44}+\frac{7\sqrt{55}Q_x^{\beta}xyz\left(2x^2-y^2-z^2\right)}{22}$$
$$-\frac{5\sqrt{33}Q_y^{\alpha}z\left(6x^4-51x^2y^2+5x^2z^2+6y^4+5y^2z^2-z^4\right)}{264}-\frac{\sqrt{55}Q_y^{\beta}z\left(10x^4+27x^2y^2-29x^2z^2-4y^4-y^2z^2+3z^4\right)}{88}$$
$$+\frac{5\sqrt{33}Q_z^{\alpha}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{264}-\frac{\sqrt{55}Q_z^{\beta}y\left(10x^4-29x^2y^2+27x^2z^2+3y^4-y^2z^2-4z^4\right)}{88}$$

$\vec{\mathbb{G}}_3^{(3,2)}[q](B_g,2)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{7\sqrt{55}Q_{xyz}z\left(x-y\right)\left(x+y\right)\left(x^2+y^2-2z^2\right)}{44}-\frac{5\sqrt{33}Q_x^{\alpha}y\left(6x^4+5x^2y^2-51x^2z^2-y^4+5y^2z^2+6z^4\right)}{264}$$
$$+\frac{\sqrt{55}Q_x^{\beta}y\left(4x^4+x^2y^2-27x^2z^2-3y^4+29y^2z^2-10z^4\right)}{88}-\frac{5\sqrt{33}Q_y^{\alpha}x\left(x^4-5x^2y^2-5x^2z^2-6y^4+51y^2z^2-6z^4\right)}{264}$$
$$-\frac{\sqrt{55}Q_y^{\beta}x\left(3x^4-x^2y^2-29x^2z^2-4y^4+27y^2z^2+10z^4\right)}{88}-\frac{7\sqrt{55}Q_z^{\beta}xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{G}}_3^{(3,2)}[q](B_g,3)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$-\frac{\sqrt{33}Q_{xyz}x\left(8x^4-40x^2y^2-40x^2z^2+15y^4+30y^2z^2+15z^4\right)}{132}-\frac{7\sqrt{55}Q_x^{\alpha}xyz\left(2x^2-y^2-z^2\right)}{22}$$
$$-\frac{\sqrt{55}Q_y^{\alpha}z\left(4x^4-27x^2y^2+x^2z^2-10y^4+29y^2z^2-3z^4\right)}{88}-\frac{\sqrt{33}Q_y^{\beta}z\left(60x^4-195x^2y^2-55x^2z^2+60y^4-55y^2z^2+11z^4\right)}{264}$$
$$-\frac{\sqrt{55}Q_z^{\alpha}y\left(4x^4+x^2y^2-27x^2z^2-3y^4+29y^2z^2-10z^4\right)}{88}+\frac{\sqrt{33}Q_z^{\beta}y\left(60x^4-55x^2y^2-195x^2z^2+11y^4-55y^2z^2+60z^4\right)}{264}$$

$\vec{\mathbb{G}}_3^{(3,2)}[q](B_g,4)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{\sqrt{33}Q_{xyz}z\left(15x^4+30x^2y^2-40x^2z^2+15y^4-40y^2z^2+8z^4\right)}{132}+\frac{\sqrt{55}Q_x^{\alpha}y\left(10x^4-29x^2y^2+27x^2z^2+3y^4-y^2z^2-4z^4\right)}{88}$$
$$-\frac{\sqrt{33}Q_x^{\beta}y\left(60x^4-55x^2y^2-195x^2z^2+11y^4-55y^2z^2+60z^4\right)}{264}+\frac{\sqrt{55}Q_y^{\alpha}x\left(3x^4-29x^2y^2-x^2z^2+10y^4+27y^2z^2-4z^4\right)}{88}$$
$$+\frac{\sqrt{33}Q_y^{\beta}x\left(11x^4-55x^2y^2-55x^2z^2+60y^4-195y^2z^2+60z^4\right)}{264}+\frac{7\sqrt{55}Q_z^{\alpha}xyz\left(x^2+y^2-2z^2\right)}{22}$$

$\vec{\mathbb{Q}}_3^{(3,-3)}[q](A_u,1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$Q_{xyz}$$

$\vec{\mathbb{Q}}_3^{(3,-3)}[q](A_u,2)$
** symmetry

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$Q_y^{\alpha}$$

$\vec{\mathbb{Q}}_3^{(3,-3)}[q](A_u,3)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$Q_y^\beta$$

$\vec{\mathbb{Q}}_3^{(3,-1)}[q](A_u, 1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-Q_x^\alpha yz - Q_y^\alpha xz - Q_z^\alpha xy$$

$\vec{\mathbb{Q}}_3^{(3,-1)}[q](A_u, 2)$
** symmetry

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

** expression

$$-Q_{xyz}xz - \frac{\sqrt{15}Q_x^\alpha xy}{20} + \frac{Q_x^\beta xy}{4} - \frac{\sqrt{15}Q_y^\alpha\left(x^2 - 2y^2 + z^2\right)}{15} + \frac{Q_y^\beta(x-z)(x+z)}{2} - \frac{\sqrt{15}Q_z^\alpha yz}{20} - \frac{Q_z^\beta yz}{4}$$

$\vec{\mathbb{Q}}_3^{(3,-1)}[q](A_u, 3)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{Q_x^\alpha xy}{4} - \frac{\sqrt{15}Q_x^\beta xy}{4} + \frac{Q_y^\alpha(x-z)(x+z)}{2} + \frac{Q_z^\alpha yz}{4} - \frac{\sqrt{15}Q_z^\beta yz}{4}$$

$\vec{\mathbb{Q}}_3^{(3,1)}[q](A_u, 1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$-\frac{7\sqrt{22}Q_{xyz}\left(x^4 - 3x^2y^2 - 3x^2z^2 + y^4 - 3y^2z^2 + z^4\right)}{66} - \frac{\sqrt{330}Q_x^\alpha yz\left(6x^2 - y^2 - z^2\right)}{132} + \frac{35\sqrt{22}Q_x^\beta yz(y-z)(y+z)}{132}$$
$$+\frac{\sqrt{330}Q_y^\alpha xz\left(x^2 - 6y^2 + z^2\right)}{132} - \frac{35\sqrt{22}Q_y^\beta xz(x-z)(x+z)}{132} + \frac{\sqrt{330}Q_z^\alpha xy\left(x^2 + y^2 - 6z^2\right)}{132} + \frac{35\sqrt{22}Q_z^\beta xy(x-y)(x+y)}{132}$$

$\vec{\mathbb{Q}}_3^{(3,1)}[q](A_u, 2)$
** symmetry

$$-\frac{3x^2y}{2} + y^3 - \frac{3yz^2}{2}$$

** expression

$$\frac{\sqrt{330}Q_{xyz}xz\left(x^2 - 6y^2 + z^2\right)}{132} - \frac{5\sqrt{22}Q_x^\alpha xy\left(x^2 + y^2 - 6z^2\right)}{44} - \frac{\sqrt{330}Q_x^\beta xy\left(9x^2 - 5y^2 - 12z^2\right)}{132} + \frac{\sqrt{22}Q_y^\alpha\left(3x^4 - 24x^2y^2 + 6x^2z^2 + 8y^4 - 24y^2z^2 + 3z^4\right)}{88}$$
$$-\frac{\sqrt{330}Q_y^\beta(x-z)(x+z)\left(x^2 - 6y^2 + z^2\right)}{264} + \frac{5\sqrt{22}Q_z^\alpha yz\left(6x^2 - y^2 - z^2\right)}{44} - \frac{\sqrt{330}Q_z^\beta yz\left(12x^2 + 5y^2 - 9z^2\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(3,1)}[q](A_u, 3)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2} + \frac{\sqrt{15}yz^2}{2}$$

** expression

$$-\frac{35\sqrt{22}Q_{xyz}xz(x-z)(x+z)}{132} - \frac{\sqrt{330}Q_x^\alpha xy\left(5x^2 - 9y^2 + 12z^2\right)}{132} - \frac{5\sqrt{22}Q_x^\beta xy\left(x^2 + y^2 - 6z^2\right)}{132} - \frac{\sqrt{330}Q_y^\alpha(x-z)(x+z)\left(x^2 - 6y^2 + z^2\right)}{264}$$
$$+\frac{7\sqrt{22}Q_y^\beta\left(x^4 + 12x^2y^2 - 18x^2z^2 - 4y^4 + 12y^2z^2 + z^4\right)}{264} + \frac{\sqrt{330}Q_z^\alpha yz\left(12x^2 - 9y^2 + 5z^2\right)}{132} + \frac{5\sqrt{22}Q_z^\beta yz\left(6x^2 - y^2 - z^2\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(3,3)}[q](A_u, 1)$
** symmetry

$$\sqrt{15}xyz$$

** expression

$$\frac{\sqrt{429}Q_{xyz}\left(2x^6-15x^4y^2-15x^4z^2-15x^2y^4+180x^2y^2z^2-15x^2z^4+2y^6-15y^4z^2-15y^2z^4+2z^6\right)}{286}$$
$$+\frac{7\sqrt{715}Q_x^{\alpha}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{572}+\frac{21\sqrt{429}Q_x^{\beta}yz\left(y-z\right)\left(y+z\right)\left(10x^2-y^2-z^2\right)}{572}$$
$$+\frac{7\sqrt{715}Q_y^{\alpha}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{572}+\frac{21\sqrt{429}Q_y^{\beta}xz\left(x-z\right)\left(x+z\right)\left(x^2-10y^2+z^2\right)}{572}$$
$$+\frac{7\sqrt{715}Q_z^{\alpha}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}-\frac{21\sqrt{429}Q_z^{\beta}xy\left(x-y\right)\left(x+y\right)\left(x^2+y^2-10z^2\right)}{572}$$

$\vec{\mathbb{Q}}_3^{(3,3)}[q](A_u,2)$
** symmetry

$$-\frac{3x^2y}{2}+y^3-\frac{3yz^2}{2}$$

** expression

$$\frac{7\sqrt{715}Q_{xyz}xz\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{572}-\frac{35\sqrt{429}Q_x^{\alpha}xy\left(2x^4-7x^2y^2+x^2z^2+2y^4+y^2z^2-z^4\right)}{1144}$$
$$+\frac{7\sqrt{715}Q_x^{\beta}xy\left(4x^4-19x^2y^2+17x^2z^2+10y^4-43y^2z^2+13z^4\right)}{1144}$$
$$-\frac{5\sqrt{429}Q_y^{\alpha}\left(5x^6-90x^4y^2+15x^4z^2+120x^2y^4-180x^2y^2z^2+15x^2z^4-16y^6+120y^4z^2-90y^2z^4+5z^6\right)}{3432}$$
$$-\frac{7\sqrt{715}Q_y^{\beta}\left(x-z\right)\left(x+z\right)\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{1144}$$
$$+\frac{35\sqrt{429}Q_z^{\alpha}yz\left(x^4-x^2y^2-x^2z^2-2y^4+7y^2z^2-2z^4\right)}{1144}-\frac{7\sqrt{715}Q_z^{\beta}yz\left(13x^4-43x^2y^2+17x^2z^2+10y^4-19y^2z^2+4z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,3)}[q](A_u,3)$
** symmetry

$$-\frac{\sqrt{15}x^2y}{2}+\frac{\sqrt{15}yz^2}{2}$$

** expression

$$\frac{21\sqrt{429}Q_{xyz}xz\left(x-z\right)\left(x+z\right)\left(x^2-10y^2+z^2\right)}{572}-\frac{7\sqrt{715}Q_x^{\alpha}xy\left(10x^4-19x^2y^2-43x^2z^2+4y^4+17y^2z^2+13z^4\right)}{1144}$$
$$+\frac{7\sqrt{429}Q_x^{\beta}xy\left(4x^4-25x^2y^2+35x^2z^2+4y^4+35y^2z^2-35z^4\right)}{1144}-\frac{7\sqrt{715}Q_y^{\alpha}\left(x-z\right)\left(x+z\right)\left(x^4-16x^2y^2+2x^2z^2+16y^4-16y^2z^2+z^4\right)}{1144}$$
$$-\frac{\sqrt{429}Q_y^{\beta}\left(13x^6-150x^4y^2-45x^4z^2+60x^2y^4+540x^2y^2z^2-45x^2z^4-8y^6+60y^4z^2-150y^2z^4+13z^6\right)}{1144}$$
$$+\frac{7\sqrt{715}Q_z^{\alpha}yz\left(13x^4+17x^2y^2-43x^2z^2+4y^4-19y^2z^2+10z^4\right)}{1144}-\frac{7\sqrt{429}Q_z^{\beta}yz\left(35x^4-35x^2y^2-35x^2z^2-4y^4+25y^2z^2-4z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,-3)}[q](B_u,1)$
** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

** expression

$$Q_x^{\alpha}$$

$\vec{\mathbb{Q}}_3^{(3,-3)}[q](B_u,2)$
** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$Q_z^{\alpha}$$

$\vec{\mathbb{Q}}_3^{(3,-3)}[q](B_u,3)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$Q_x^{\beta}$$

$\vec{\mathbb{Q}}_3^{(3,-3)}[q](B_u,4)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$Q_z^\beta$$

$\vec{\mathbb{Q}}_3^{(3,-1)}[q](B_u, 1)$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$-Q_{xyz}yz + \frac{\sqrt{15}Q_x^\alpha\left(2x^2 - y^2 - z^2\right)}{15} - \frac{Q_x^\beta(y-z)(y+z)}{2} - \frac{\sqrt{15}Q_y^\alpha xy}{20} - \frac{Q_y^\beta xy}{4} - \frac{\sqrt{15}Q_z^\alpha xz}{20} + \frac{Q_z^\beta xz}{4}$$

$\vec{\mathbb{Q}}_3^{(3,-1)}[q](B_u, 2)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$-Q_{xyz}xy - \frac{\sqrt{15}Q_x^\alpha xz}{20} - \frac{Q_x^\beta xz}{4} - \frac{\sqrt{15}Q_y^\alpha yz}{20} + \frac{Q_y^\beta yz}{4} - \frac{\sqrt{15}Q_z^\alpha\left(x^2 + y^2 - 2z^2\right)}{15} - \frac{Q_z^\beta(x-y)(x+y)}{2}$$

$\vec{\mathbb{Q}}_3^{(3,-1)}[q](B_u, 3)$
** symmetry

$$\frac{\sqrt{15}xy^2}{2} - \frac{\sqrt{15}xz^2}{2}$$

** expression

$$-\frac{Q_x^\alpha(y-z)(y+z)}{2} + \frac{Q_y^\alpha xy}{4} - \frac{\sqrt{15}Q_y^\beta xy}{4} - \frac{Q_z^\alpha xz}{4} - \frac{\sqrt{15}Q_z^\beta xz}{4}$$

$\vec{\mathbb{Q}}_3^{(3,-1)}[q](B_u, 4)$
** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{Q_x^\alpha xz}{4} - \frac{\sqrt{15}Q_x^\beta xz}{4} - \frac{Q_y^\alpha yz}{4} - \frac{\sqrt{15}Q_y^\beta yz}{4} - \frac{Q_z^\alpha(x-y)(x+y)}{2}$$

$\vec{\mathbb{Q}}_3^{(3,1)}[q](B_u, 1)$
** symmetry

$$x^3 - \frac{3xy^2}{2} - \frac{3xz^2}{2}$$

** expression

$$-\frac{\sqrt{330}Q_{xyz}yz\left(6x^2 - y^2 - z^2\right)}{132} + \frac{\sqrt{22}Q_x^\alpha\left(8x^4 - 24x^2y^2 - 24x^2z^2 + 3y^4 + 6y^2z^2 + 3z^4\right)}{88} - \frac{\sqrt{330}Q_x^\beta(y-z)(y+z)\left(6x^2 - y^2 - z^2\right)}{264}$$
$$- \frac{5\sqrt{22}Q_y^\alpha xy\left(x^2 + y^2 - 6z^2\right)}{44} - \frac{\sqrt{330}Q_y^\beta xy\left(5x^2 - 9y^2 + 12z^2\right)}{132} - \frac{5\sqrt{22}Q_z^\alpha xz\left(x^2 - 6y^2 + z^2\right)}{44} + \frac{\sqrt{330}Q_z^\beta xz\left(5x^2 + 12y^2 - 9z^2\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(3,1)}[q](B_u, 2)$
** symmetry

$$-\frac{3x^2z}{2} - \frac{3y^2z}{2} + z^3$$

** expression

$$\frac{\sqrt{330}Q_{xyz}xy\left(x^2 + y^2 - 6z^2\right)}{132} - \frac{5\sqrt{22}Q_x^\alpha xz\left(x^2 - 6y^2 + z^2\right)}{44} + \frac{\sqrt{330}Q_x^\beta xz\left(9x^2 - 12y^2 - 5z^2\right)}{132} + \frac{5\sqrt{22}Q_y^\alpha yz\left(6x^2 - y^2 - z^2\right)}{44}$$
$$+ \frac{\sqrt{330}Q_y^\beta yz\left(12x^2 - 9y^2 + 5z^2\right)}{132} + \frac{\sqrt{22}Q_z^\alpha\left(3x^4 + 6x^2y^2 - 24x^2z^2 + 3y^4 - 24y^2z^2 + 8z^4\right)}{88} + \frac{\sqrt{330}Q_z^\beta(x-y)(x+y)\left(x^2 + y^2 - 6z^2\right)}{264}$$

$\vec{\mathbb{Q}}_3^{(3,1)}[q](B_u, 3)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$\frac{35\sqrt{22}Q_{xyz}yz\left(y-z\right)\left(y+z\right)}{132}-\frac{\sqrt{330}Q_x^{\alpha}\left(y-z\right)\left(y+z\right)\left(6x^2-y^2-z^2\right)}{264}-\frac{7\sqrt{22}Q_x^{\beta}\left(4x^4-12x^2y^2-12x^2z^2-y^4+18y^2z^2-z^4\right)}{264}$$
$$-\frac{\sqrt{330}Q_y^{\alpha}xy\left(9x^2-5y^2-12z^2\right)}{132}-\frac{5\sqrt{22}Q_y^{\beta}xy\left(x^2+y^2-6z^2\right)}{132}+\frac{\sqrt{330}Q_z^{\alpha}xz\left(9x^2-12y^2-5z^2\right)}{132}-\frac{5\sqrt{22}Q_z^{\beta}xz\left(x^2-6y^2+z^2\right)}{132}$$

$\vec{\mathbb{Q}}_3^{(3,1)}[q](B_u,4)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2}-\frac{\sqrt{15}y^2z}{2}$$

** expression

$$\frac{35\sqrt{22}Q_{xyz}xy\left(x-y\right)\left(x+y\right)}{132}+\frac{\sqrt{330}Q_x^{\alpha}xz\left(5x^2+12y^2-9z^2\right)}{132}-\frac{5\sqrt{22}Q_x^{\beta}xz\left(x^2-6y^2+z^2\right)}{132}-\frac{\sqrt{330}Q_y^{\alpha}yz\left(12x^2+5y^2-9z^2\right)}{132}$$
$$+\frac{5\sqrt{22}Q_y^{\beta}yz\left(6x^2-y^2-z^2\right)}{132}+\frac{\sqrt{330}Q_z^{\alpha}\left(x-y\right)\left(x+y\right)\left(x^2+y^2-6z^2\right)}{264}+\frac{7\sqrt{22}Q_z^{\beta}\left(x^4-18x^2y^2+12x^2z^2+y^4+12y^2z^2-4z^4\right)}{264}$$

$\vec{\mathbb{Q}}_3^{(3,3)}[q](B_u,1)$

** symmetry

$$x^3-\frac{3xy^2}{2}-\frac{3xz^2}{2}$$

** expression

$$\frac{7\sqrt{715}Q_{xyz}yz\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{572}$$
$$+\frac{5\sqrt{429}Q_x^{\alpha}\left(16x^6-120x^4y^2-120x^4z^2+90x^2y^4+180x^2y^2z^2+90x^2z^4-5y^6-15y^4z^2-15y^2z^4-5z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}Q_x^{\beta}\left(y-z\right)\left(y+z\right)\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}Q_y^{\alpha}xy\left(2x^4-7x^2y^2+x^2z^2+2y^4+y^2z^2-z^4\right)}{1144}-\frac{7\sqrt{715}Q_y^{\beta}xy\left(10x^4-19x^2y^2-43x^2z^2+4y^4+17y^2z^2+13z^4\right)}{1144}$$
$$-\frac{35\sqrt{429}Q_z^{\alpha}xz\left(2x^4+x^2y^2-7x^2z^2-y^4+y^2z^2+2z^4\right)}{1144}+\frac{7\sqrt{715}Q_z^{\beta}xz\left(10x^4-43x^2y^2-19x^2z^2+13y^4+17y^2z^2+4z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,3)}[q](B_u,2)$

** symmetry

$$-\frac{3x^2z}{2}-\frac{3y^2z}{2}+z^3$$

** expression

$$\frac{7\sqrt{715}Q_{xyz}xy\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{572}-\frac{35\sqrt{429}Q_x^{\alpha}xz\left(2x^4+x^2y^2-7x^2z^2-y^4+y^2z^2+2z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}Q_x^{\beta}xz\left(4x^4+17x^2y^2-19x^2z^2+13y^4-43y^2z^2+10z^4\right)}{1144}$$
$$+\frac{35\sqrt{429}Q_y^{\alpha}yz\left(x^4-x^2y^2-x^2z^2-2y^4+7y^2z^2-2z^4\right)}{1144}+\frac{7\sqrt{715}Q_y^{\beta}yz\left(13x^4+17x^2y^2-43x^2z^2+4y^4-19y^2z^2+10z^4\right)}{1144}$$
$$-\frac{5\sqrt{429}Q_z^{\alpha}\left(5x^6+15x^4y^2-90x^4z^2+15x^2y^4-180x^2y^2z^2+120x^2z^4+5y^6-90y^4z^2+120y^2z^4-16z^6\right)}{3432}$$
$$+\frac{7\sqrt{715}Q_z^{\beta}\left(x-y\right)\left(x+y\right)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,3)}[q](B_u,3)$

** symmetry

$$\frac{\sqrt{15}xy^2}{2}-\frac{\sqrt{15}xz^2}{2}$$

** expression

$$\frac{21\sqrt{429}Q_{xyz}yz\left(y-z\right)\left(y+z\right)\left(10x^2-y^2-z^2\right)}{572}+\frac{7\sqrt{715}Q_x^{\alpha}\left(y-z\right)\left(y+z\right)\left(16x^4-16x^2y^2-16x^2z^2+y^4+2y^2z^2+z^4\right)}{1144}$$
$$+\frac{\sqrt{429}Q_x^{\beta}\left(8x^6-60x^4y^2-60x^4z^2+150x^2y^4-540x^2y^2z^2+150x^2z^4-13y^6+45y^4z^2+45y^2z^4-13z^6\right)}{1144}$$
$$+\frac{7\sqrt{715}Q_y^{\alpha}xy\left(4x^4-19x^2y^2+17x^2z^2+10y^4-43y^2z^2+13z^4\right)}{1144}+\frac{7\sqrt{429}Q_y^{\beta}xy\left(4x^4-25x^2y^2+35x^2z^2+4y^4+35y^2z^2-35z^4\right)}{1144}$$
$$-\frac{7\sqrt{715}Q_z^{\alpha}xz\left(4x^4+17x^2y^2-19x^2z^2+13y^4-43y^2z^2+10z^4\right)}{1144}+\frac{7\sqrt{429}Q_z^{\beta}xz\left(4x^4+35x^2y^2-25x^2z^2-35y^4+35y^2z^2+4z^4\right)}{1144}$$

$\vec{\mathbb{Q}}_3^{(3,3)}[q](B_u, 4)$

** symmetry

$$\frac{\sqrt{15}x^2z}{2} - \frac{\sqrt{15}y^2z}{2}$$

** expression

$$-\frac{21\sqrt{429}Q_{xyz}xy\,(x-y)\,(x+y)\left(x^2+y^2-10z^2\right)}{572} + \frac{7\sqrt{715}Q_x^{\alpha}xz\left(10x^4-43x^2y^2-19x^2z^2+13y^4+17y^2z^2+4z^4\right)}{1144}$$
$$+\frac{7\sqrt{429}Q_x^{\beta}xz\left(4x^4+35x^2y^2-25x^2z^2-35y^4+35y^2z^2+4z^4\right)}{1144} - \frac{7\sqrt{715}Q_y^{\alpha}yz\left(13x^4-43x^2y^2+17x^2z^2+10y^4-19y^2z^2+4z^4\right)}{1144}$$
$$-\frac{7\sqrt{429}Q_y^{\beta}yz\left(35x^4-35x^2y^2-35x^2z^2-4y^4+25y^2z^2-4z^4\right)}{1144} + \frac{7\sqrt{715}Q_z^{\alpha}\,(x-y)\,(x+y)\left(x^4+2x^2y^2-16x^2z^2+y^4-16y^2z^2+16z^4\right)}{1144}$$
$$-\frac{\sqrt{429}Q_z^{\beta}\left(13x^6-45x^4y^2-150x^4z^2-45x^2y^4+540x^2y^2z^2+60x^2z^4+13y^6-150y^4z^2+60y^2z^4-8z^6\right)}{1144}$$